

— A FIELD GUIDE —

# Nine **RAG** Architectures, Compared.

From Standard to Multimodal — a practitioner's guide to when each architecture earns its place in production.

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# Standard RAG

## ● PIPELINE

01

**User Query**

02

**Query Processing**

03

**Retrieval**

04

**Document Selection**

05

**Context Integration**

06

**LLM Response**

## ★ TIP

Start with smaller chunk sizes (256–512 tokens) and experiment with overlap to improve coverage without overwhelming the context window.

## ► WHEN TO USE

Best for straightforward fact retrieval where accuracy matters but complexity is low.

## ◆ EXAMPLE

FAQ bots that surface up-to-date company policies or knowledge base entries.



# DeepRAG

Hierarchical + Process Supervision

## ● PIPELINE

- 01 User Query
- 02 Question Decomposition
- 03 Retrieval per sub-question
- 04 Reasoning + Verification
- 05 Final Response

## ★ TIP

Use hierarchical indicators to manage nested dependencies, and combine process supervision (reward signals) to align sub-queries with final answers.

## ► WHEN TO USE

Multi-hop or domain-specific (e.g. biomedical) questions needing structured decomposition.

## ◆ EXAMPLE

Biomedical QA over linked concept databases (e.g. UMLS).





# MA-RAG

Multi-Agent RAG

## • PIPELINE

01

User Query

02

Planner + Extractor + Retriever

03

Combined Retrieval & Reasoning

04

LLM Output

## ★ TIP

Each agent handles a subtask, and only the necessary agents are invoked per query. Agents use chain-of-thought prompting for interpretability.

## ► WHEN TO USE

Ambiguous or complex queries where decomposition and agent collaboration helps.

## ◆ EXAMPLE

Open-domain QA across heterogeneous sources.





# Corrective RAG

## ● PIPELINE

- 01 User Query
- 02 Initial Response
- 03 Error Detection
- 04 Retrieval
- 05 Response Correction
- 06 Final Response

## ★ TIP

Use a lightweight verification loop — either a separate classifier or an auxiliary LLM — to flag uncertain segments and trigger corrective retrieval.

## ► WHEN TO USE

Suitable when outputs must be double-checked before being shown to the user.

## ◆ EXAMPLE

Clinical decision-support assistants that cross-verify against trusted medical literature.



# Speculative RAG

## ● PIPELINE

- 01 User Query
- 02 Small Model Draft
- 03 Parallel Retrievals
- 04 Large Model Verification
- 05 Final Response

## ★ TIP

Pair a smaller, faster model for drafting with a larger model that validates against retrieved documents; adjust retrieval parallelism based on latency budgets.

## ► WHEN TO USE

Ideal when response speed is crucial but correctness can't be sacrificed.

## ◆ EXAMPLE

Live chat agents for e-commerce that need rapid draft answers while verifying product details.





# Fusion RAG

## ● PIPELINE

01

**User Query**

02

**Multiple Query Generation**

03

**Multi-Retrieval**

04

**Fusion of Documents**

05

**LLM Response**

## ★ TIP

Employ a fusion method like Reciprocal Rank Fusion to merge results, plus weighting sources according to reliability and domain relevance.

## ► WHEN TO USE

Useful when queries require pulling evidence from different types of sources or formats.

## ◆ EXAMPLE

Academic research assistants synthesizing across journal articles, patents, and technical blogs.





# RAG-Gym

Process-Supervised Agentic RAG

## ● PIPELINE

01

User Query

02

Agentic Planning

03

Step-by-step Retrieval

04

LLM Generation + Supervision

## ★ TIP

Use process-level rewards to train agents — reward which retrieval or reformulation step is better, not just the final answer. The RAG-Gym / ReSearch architecture is from a recent arXiv paper.

## ► WHEN TO USE

For multi-step reasoning or tasks that benefit from supervising intermediate steps.

## ◆ EXAMPLE

QA systems that have to plan a search path rather than one-shot retrieval.





# Modular RAG

## ● PIPELINE

01

**User Query**

02

**Module Routing**

03

**Retrieval / Rewriting / Fusion**

04

**LLM Generation**

## ★ TIP

Decompose into operators (routing, query rewriting, fusion) so you can replace your reranker or reformulator independently. The Modular RAG paper identifies linear, conditional, branching, and looping patterns.

## ► WHEN TO USE

When you want flexibility in customizing or swapping parts of the RAG pipeline.

## ◆ EXAMPLE

Research systems exploring new retrieval/fusion strategies.





# SAM-RAG

Self-adaptive Multimodal RAG

## ● PIPELINE

01

Multi-modal Query

02

Adaptive Cross-Modal Retrieval

03

Filtering & Quality Check

04

LLM Response

## ★ TIP

Dynamically select which modalities to retrieve from, verify the relevance of retrieved multimodal data, and integrate them in context. The model adapts retrieval count and modality per query.

## ► WHEN TO USE

When queries or data involve images, audio, or mixed modalities.

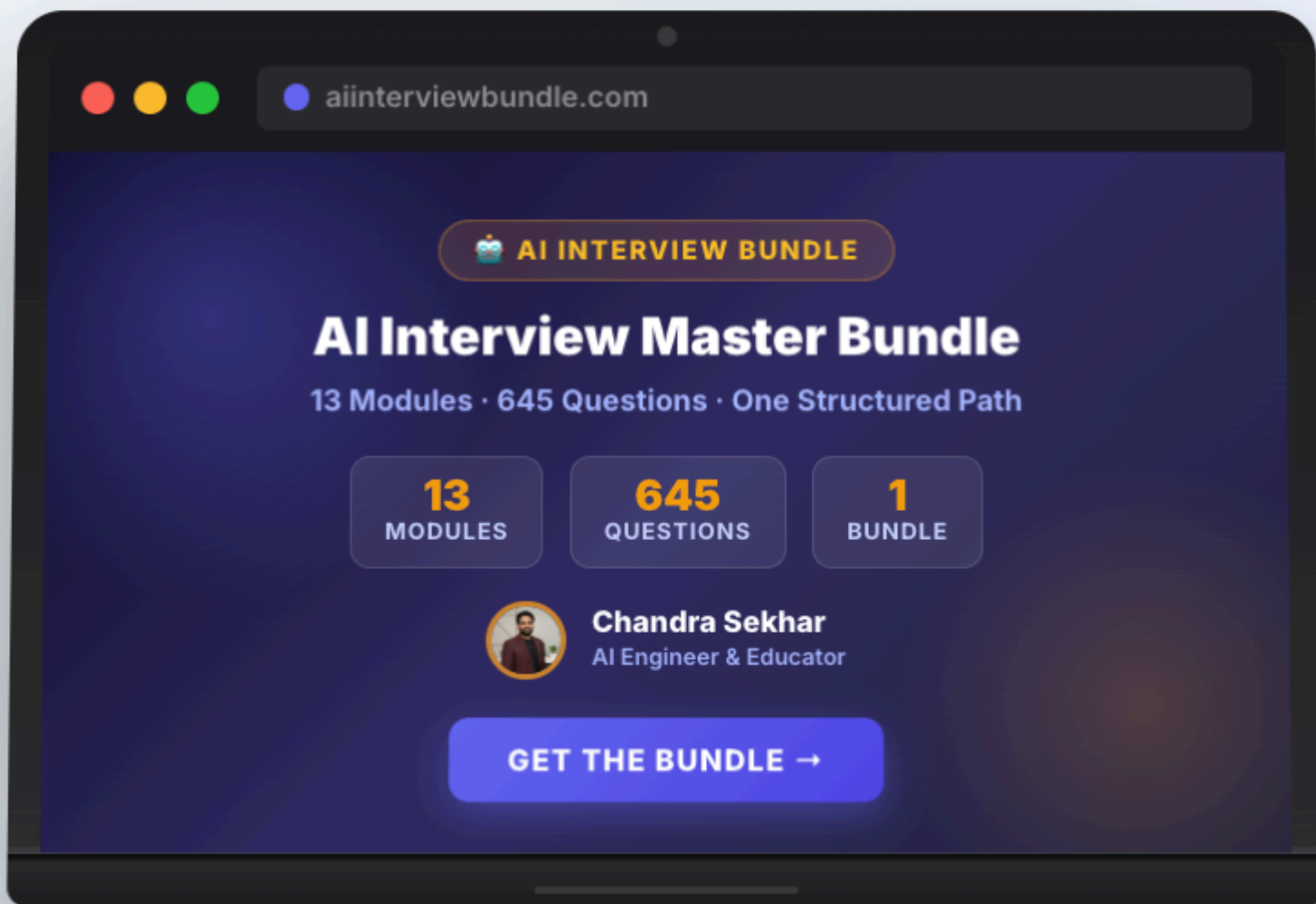
## ◆ EXAMPLE

Q&A combining text + images (e.g. product descriptions + photos).



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